TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	90.7 ± 0.78
Correlation Stability	93.4 ± 0.66
Distribution Consistency	91.6 ± 1.02
Entropy Preservation	95.2 ± 0.75
Mean Z Score Drift	0.1 ± 0.01
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	99.5 ± 0.50
Feature Accuracy	94.7 ± 1.10
Novelty Score	0.8 ± 0.01

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Dataset Audit: heart_disease.csv

1. Dataset Portrait

Attribute	Value
Rows	303
Columns	14
Numeric features	14
Categorical features	0
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
age	54.439	53.314	-1.125	2.1%	OK
sex	0.680	0.694	+0.014	2.1%	OK
cp	3.158	3.145	-0.014	0.4%	OK
trestbps	131.690	131.066	-0.624	0.5%	OK
chol	246.693	245.926	-0.767	0.3%	OK
fbs	0.149	0.120	-0.029	19.3%	Warn
restecg	0.990	1.054	+0.064	6.4%	OK
thalach	149.607	151.322	+1.715	1.1%	OK
exang	0.327	0.318	-0.009	2.6%	OK
oldpeak	1.040	0.935	-0.105	10.1%	Warn
slope	1.601	1.603	+0.003	0.2%	OK
ca	0.672	0.628	-0.044	6.6%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	90.7 ± 0.78
Correlation Stability (*2)	93.4 ± 0.66
Distribution Consistency (*3)	91.6 ± 1.02
Entropy Preservation (*4)	95.2 ± 0.75
Mean Z Score Drift (*5)	0.1 ± 0.01
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.0 ± 0.00

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	99.5 ± 0.50
Feature Accuracy (*10)	94.7 ± 1.10
Novelty Score (*11)	0.8 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5426229508196723	0.5672131147540983	0.5819672131147542	0.03934426229508194	
F1	0.5198013329305267	0.5287757053862534	0.5599660518607866	0.04016471893025985	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5770491803278688	0.5852459016393442	0.5803278688524591	0.00327868852459016	
F1	0.5374233649387901	0.5220576317474851	0.5288506566036199	-0.00857270833517019	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5573770491803279	0.5459016393442623	0.5918032786885246	0.03442622950819673	
F1	0.568660979990255	0.5435747475894883	0.592227133010329	0.02356615302007406	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5540983606557377	0.537704918032787	0.5245901639344263	-0.02950819672131476	*
F1	0.44422020714365684	0.41421005791945475	0.4166154937290306	-0.02760471341462617	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5475409836065573	0.5475409836065573	0.5475409836065573	0	
F1	0.3899270600110827	0.3899270600110827	0.3899270600110827	0	

Model: AB

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6081967213114754	0.5557377049180328	0.6180327868852461	0.009836065573770491	
F1	0.6070236204175147	0.5253279104523846	0.5930772810258199	-0.013946339391694706	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5377049180327869	0.5737704918032788	0.5852459016393443	0.04754098360655738	
F1	0.5239891946681083	0.5399943556534478	0.5625605669077498	0.03857137223964144	*

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5836065573770493	0.5409836065573771	0.5934426229508196	0.009836065573770475	
F1	0.5463850924114617	0.5089668708929875	0.5515610028347725	0.005175910423310737	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5295081967213114	0.5098360655737705	0.5032786885245901	-0.026229508196721298	
F1	0.5331511979212488	0.5039341746275512	0.5023146555368924	-0.030836542384356326	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5983606557377049	0.5967213114754099	0.6032786885245901	0.004918032786885235	
F1	0.5648347697855012	0.5410590601033307	0.553336239872063	-0.011498529913438166	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	63.0	1.0	1.0	145.0	233.0	1.0	2.0	150.0	0.0	2.3
2	67.0	1.0	4.0	160.0	286.0	0.0	2.0	108.0	1.0	1.5
3	67.0	1.0	4.0	120.0	229.0	0.0	2.0	129.0	1.0	2.6
4	37.0	1.0	3.0	130.0	250.0	0.0	0.0	187.0	0.0	3.5
5	41.0	0.0	2.0	130.0	204.0	0.0	2.0	172.0	0.0	1.4
6	56.0	1.0	2.0	120.0	236.0	0.0	0.0	178.0	0.0	0.8
7	62.0	0.0	4.0	140.0	268.0	0.0	2.0	160.0	0.0	3.6
8	57.0	0.0	4.0	120.0	354.0	0.0	0.0	163.0	1.0	0.6
9	63.0	1.0	4.0	130.0	254.0	0.0	2.0	147.0	0.0	1.4
10	53.0	1.0	4.0	140.0	203.0	1.0	2.0	155.0	1.0	3.1
11	57.0	1.0	4.0	140.0	192.0	0.0	0.0	148.0	0.0	0.4
12	56.0	0.0	2.0	140.0	294.0	0.0	2.0	153.0	0.0	1.3
13	56.0	1.0	3.0	130.0	256.0	1.0	2.0	142.0	1.0	0.6
14	44.0	1.0	2.0	120.0	263.0	0.0	0.0	173.0	0.0	0.0
15	52.0	1.0	3.0	172.0	199.0	1.0	0.0	162.0	0.0	0.5
16	57.0	1.0	3.0	150.0	168.0	0.0	0.0	174.0	0.0	1.6
17	48.0	1.0	2.0	110.0	229.0	0.0	0.0	168.0	0.0	1.0
18	54.0	1.0	4.0	140.0	239.0	0.0	0.0	160.0	0.0	1.2
19	48.0	0.0	3.0	130.0	275.0	0.0	0.0	139.0	0.0	0.2
20	49.0	1.0	2.0	130.0	266.0	0.0	0.0	171.0	0.0	0.6

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	62.0	1.0	4.0	140.0	256.0	1.0	0.0	160.0	0.0	0.0
2	59.0	0.0	4.0	130.0	174.0	0.0	2.0	133.0	0.0	0.8
3	41.0	0.0	4.0	142.0	161.0	0.0	0.0	160.0	1.0	0.8
4	46.0	1.0	2.0	130.0	166.0	0.0	0.0	169.0	0.0	0.0
5	38.0	1.0	3.0	112.0	234.0	0.0	2.0	163.0	0.0	1.0
6	43.0	0.0	4.0	130.0	268.0	0.0	2.0	158.0	0.0	4.895656250905608
7	49.0	0.0	4.0	130.0	269.0	0.0	0.0	163.0	0.0	0.0
8	54.0	1.0	3.0	120.0	258.0	0.0	2.0	147.0	0.0	0.4
9	55.0	1.0	2.0	130.0	262.0	0.0	0.0	155.0	0.0	0.0
10	46.0	1.0	2.0	101.0	197.0	1.0	0.0	156.0	0.0	0.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	45.0	1.0	3.0	111.0	271.0	0.0	0.0	169.0	0.0	0.0
2	47.0	0.0	2.0	130.0	269.0	0.0	0.0	166.0	0.0	0.2
3	52.0	1.0	2.0	140.0	195.0	0.0	2.0	152.0	0.0	0.0
4	60.0	0.0	4.0	110.0	208.0	0.0	0.0	146.0	0.0	1.0
5	49.0	1.0	2.0	100.0	223.0	0.0	0.0	131.0	0.0	0.0
6	54.0	1.0	2.0	135.0	232.0	0.0	2.0	170.0	0.0	0.0
7	63.0	0.0	3.0	135.0	252.0	0.0	2.0	172.0	0.0	0.0
8	60.0	1.0	4.0	130.0	253.0	0.0	0.0	144.0	1.0	1.4
9	58.0	0.0	2.0	136.0	319.0	1.0	2.0	152.0	0.0	0.0
10	59.0	0.0	4.0	174.0	249.0	0.0	0.0	143.0	1.0	0.0

B.3 Audit Trace: Fold 3 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	60.0	1.0	4.0	128.0	232.0	0.0	0.0	150.0	1.0	0.1
2	54.0	1.0	4.0	112.0	219.0	0.0	0.0	146.0	1.0	0.035291084308163 08
3	52.0	0.0	3.0	110.0	201.0	0.0	2.0	130.0	0.0	0.2
4	60.0	0.0	3.0	122.0	269.0	0.0	0.0	125.0	0.0	0.0
5	56.0	1.0	2.0	120.0	199.0	0.0	2.0	179.0	0.0	0.4
6	61.0	1.0	4.0	130.0	212.0	1.0	2.0	123.0	0.0	0.1
7	49.0	1.0	3.0	120.0	188.0	0.0	0.0	139.0	0.0	2.0
8	45.0	1.0	4.0	115.0	260.0	0.0	2.0	185.0	0.0	0.0
9	45.0	0.0	2.0	130.0	234.0	0.0	2.0	175.0	0.0	0.6
10	46.0	1.0	3.0	150.0	231.0	0.0	0.0	147.0	0.0	3.6

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B.4 Audit Trace: Fold 4 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	58.0	0.0	4.0	146.0	292.0	1.0	2.0	147.0	0.0	2.2
2	68.0	1.0	4.0	130.0	343.0	0.0	2.0	169.0	0.0	0.6
3	42.0	1.0	1.0	150.0	240.0	0.0	2.0	193.0	0.0	0.0
4	69.0	0.0	3.0	110.0	322.0	0.0	0.0	147.0	0.0	1.2
5	67.0	1.0	3.0	120.0	213.0	0.0	0.0	126.0	1.0	1.9
6	52.0	1.0	2.0	138.0	354.0	0.0	2.0	143.0	0.0	1.4
7	57.0	1.0	4.0	140.0	192.0	0.0	0.0	148.0	0.0	0.4
8	63.0	0.0	4.0	150.0	407.0	0.0	2.0	154.0	0.0	4.0
9	57.0	0.0	4.0	128.0	303.0	0.0	2.0	159.0	0.0	0.0
10	60.0	1.0	4.0	117.0	230.0	1.0	0.0	160.0	1.0	1.4

B.5 Audit Trace: Fold 5 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	62.0	1.0	4.0	150.0	258.0	0.0	2.0	157.0	0.0	0.12824107958558245
2	45.0	0.0	3.0	102.0	269.0	0.0	0.0	158.0	0.0	0.0
3	75.0	1.0	4.0	150.0	265.0	0.0	2.0	113.0	1.0	3.9018012578433514
4	59.0	0.0	4.0	112.0	212.0	1.0	0.0	148.0	0.0	0.8
5	43.0	1.0	4.0	125.0	257.0	0.0	2.0	161.0	0.0	0.6
6	57.0	1.0	3.0	138.0	244.0	0.0	2.0	156.0	0.0	0.0
7	50.0	0.0	2.0	120.0	244.0	0.0	0.0	162.0	0.0	1.1
8	56.0	1.0	2.0	120.0	236.0	0.0	0.0	178.0	0.0	0.8
9	61.0	1.0	3.0	150.0	243.0	1.0	0.0	137.0	1.0	1.0
10	64.0	0.0	3.0	140.0	313.0	0.0	0.0	133.0	0.0	0.2

B.6 Audit Trace: Fold 6 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	63.0	0.0	3.0	134.0	282.0	0.0	2.0	151.0	0.0	2.6
2	59.0	1.0	3.0	156.0	304.0	0.0	2.0	132.0	1.0	1.8025970046742543
3	74.0	0.0	4.0	134.0	249.0	0.0	2.0	163.0	0.0	1.6
4	53.0	0.0	1.0	140.0	277.0	0.0	0.0	153.0	0.0	0.2
5	57.0	0.0	4.0	120.0	178.0	0.0	0.0	125.0	0.0	2.3297895850503532
6	53.0	1.0	3.0	140.0	261.0	0.0	2.0	165.0	0.0	0.0
7	39.0	1.0	4.0	118.0	219.0	0.0	0.0	140.0	0.0	1.2
8	62.0	0.0	4.0	138.0	294.0	1.0	0.0	106.0	0.0	1.9
9	66.0	1.0	4.0	160.0	228.0	0.0	2.0	138.0	0.0	2.3
10	39.0	1.0	3.0	140.0	321.0	0.0	2.0	182.0	0.0	0.0

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	62.0	0.0	3.0	120.0	288.0	0.0	2.0	170.0	0.0	0.0
2	58.0	0.0	3.0	140.0	256.0	0.0	0.0	192.0	0.0	0.0
3	57.0	0.0	2.0	141.0	263.0	0.0	2.0	178.0	0.0	0.2
4	49.0	0.0	1.0	125.0	317.0	0.0	2.0	165.0	0.0	0.0
5	52.0	1.0	3.0	130.0	198.0	0.0	2.0	166.0	0.0	0.790617888660674
6	54.0	1.0	2.0	140.0	211.0	1.0	2.0	169.0	0.0	0.0
7	56.0	0.0	4.0	134.0	409.0	0.0	2.0	150.0	1.0	1.9
8	58.0	1.0	3.0	105.0	240.0	0.0	2.0	154.0	1.0	0.6
9	54.0	1.0	2.0	192.0	283.0	0.0	2.0	195.0	0.0	0.0
10	57.0	1.0	4.0	110.0	201.0	0.0	0.0	126.0	1.0	1.5

B.8 Audit Trace: Fold 8 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	54.0	1.0	4.0	120.0	177.0	0.0	0.0	132.0	0.0	1.2
2	45.0	1.0	4.0	145.0	201.0	0.0	2.0	144.0	1.0	4.153056058595604
3	61.0	0.0	4.0	126.0	259.0	0.0	0.0	156.0	0.0	0.8
4	71.0	0.0	4.0	178.0	361.0	0.0	2.0	148.0	0.0	2.4239024848769106
5	62.0	1.0	3.0	125.0	270.0	0.0	0.0	109.0	1.0	2.0
6	52.0	0.0	4.0	120.0	253.0	0.0	0.0	173.0	0.0	0.0
7	52.0	1.0	1.0	118.0	186.0	0.0	2.0	190.0	0.0	0.0
8	54.0	0.0	3.0	135.0	304.0	1.0	0.0	170.0	0.0	0.0
9	61.0	1.0	4.0	140.0	207.0	0.0	2.0	138.0	1.0	1.9
10	61.0	1.0	3.0	150.0	243.0	1.0	0.0	137.0	1.0	1.0

B.9 Audit Trace: Fold 9 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	59.0	1.0	4.0	130.0	171.0	0.0	2.0	114.0	0.0	1.0
2	57.0	1.0	3.0	140.0	157.0	0.0	0.0	120.0	0.0	0.8
3	60.0	1.0	3.0	138.0	392.0	0.0	1.0	150.0	0.0	0.0
4	61.0	0.0	4.0	126.0	206.0	0.0	2.0	109.0	1.0	2.8
5	58.0	1.0	4.0	108.0	303.0	0.0	2.0	127.0	0.0	1.0
6	48.0	0.0	3.0	100.0	232.0	0.0	0.0	165.0	0.0	2.4
7	44.0	0.0	3.0	118.0	242.0	0.0	0.0	149.0	0.0	0.3
8	38.0	1.0	1.0	120.0	231.0	0.0	0.0	182.0	1.0	3.8
9	49.0	1.0	3.0	120.0	188.0	0.0	0.0	139.0	0.0	2.0
10	61.0	1.0	4.0	120.0	260.0	0.0	0.0	140.0	1.0	3.6

B.10 Audit Trace: Fold 10 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	64.0	1.0	3.0	140.0	295.0	0.0	1.0	140.0	1.0	0.6
2	57.0	1.0	2.0	120.0	195.0	0.0	2.0	168.0	0.0	1.2319643283073531
3	41.0	1.0	3.0	130.0	253.0	1.0	2.0	178.0	1.0	0.0
4	48.0	1.0	3.0	118.0	282.0	0.0	2.0	137.0	0.0	1.0
5	46.0	1.0	4.0	120.0	241.0	0.0	2.0	162.0	1.0	0.6
6	58.0	0.0	4.0	130.0	220.0	0.0	2.0	155.0	0.0	3.0
7	63.0	0.0	3.0	135.0	252.0	0.0	2.0	172.0	0.0	0.0
8	63.0	1.0	1.0	145.0	233.0	1.0	2.0	150.0	0.0	2.3
9	60.0	1.0	4.0	125.0	258.0	0.0	2.0	141.0	1.0	2.8
10	62.0	0.0	4.0	140.0	268.0	0.0	2.0	160.0	0.0	3.6